

MASK WHAT MATTERS: CONTROLLABLE TEXT-GUIDED MASKING FOR SELF-SUPERVISED MEDICAL IMAGE ANALYSIS

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ABSTRACT

The limited availability of expert annotations in specialized domains, including medical imaging, constrains the optimization and validation of robust vision models. Self-supervised masked image modeling (MIM) mitigates this dependency by pretraining on unlabeled images, but most existing methods instantiate masking as random, high-ratio patch removal, which is computationally inefficient and weakly aligned with task semantics. Existing region-aware variants further rely on reconstruction-error heuristics or supervised localization signals, thereby restricting their transferability across tasks and imaging modalities. We introduce Mask What Matters, a controllable text-guided masking framework for self-supervised medical image analysis. The method uses vision-language models to obtain prompt-conditioned localization cues and applies region-dependent masking ratios that preferentially target diagnostically relevant structures while reducing redundant masking in background regions. This design explicitly couples the masking distribution to semantic regions of interest, improving representation learning and cross-task generalization. Across brain MRI, chest CT, and lung X-ray benchmarks, Mask What Matters consistently improves over existing MIM methods (e.g., SparK), with gains of up to +3.1 percentage points in classification accuracy, +1.3 in box average precision (BoxAP), and +1.1 in mask average precision (MaskAP) for detection. These improvements are obtained with substantially lower overall masking ratios (e.g., 40% vs. 70%), indicating that semantic control can reduce unnecessary reconstruction burden. The results establish controllable, text-conditioned masking as an effective mechanism for semantically aligned self-supervised pretraining in medical image analysis.

1 INTRODUCTION

Image classification, segmentation, and object detection constitute core prediction problems in computer vision. Recent progress in these tasks has been driven primarily by deep learning systems trained with large annotated corpora. In specialized application domains such as medical image analysis, however, high-quality annotations require expert labor, quality control, and domain-specific labeling protocols; consequently, annotation scarcity remains a central bottleneck for deploying high-capacity vision models.

Self-supervised learning (SSL) addresses this bottleneck by using unlabeled images for representation pretraining, thereby reducing dependence on manual annotations [Jaiswal et al. \(2020\)](#); [Liu et al. \(2021\)](#); [Krishnan et al. \(2022\)](#); [Huang et al. \(2023\)](#). Within SSL, MIM has become a widely used generative pretraining objective. A MIM model observes a partially masked image and is trained to reconstruct the missing content from visible context, which induces representations that encode structural regularities and semantic dependencies. This paradigm has produced strong transfer performance across downstream tasks and has been adapted to medical image analysis [He et al. \(2022\)](#); [Gupta et al. \(2025\)](#); [Xiao et al. \(2023\)](#).

Prevailing MIM methods, including masked autoencoders (MAE) [He et al. \(2022\)](#) and SparK [Tian et al. \(2023\)](#), generally sample masks at random and therefore do not explicitly condition the masking distribution on downstream regions of interest. This task-agnostic formulation has two direct

consequences: (1) the resulting representations can be semantically misaligned with downstream objectives, which reduces transfer effectiveness; and (2) to increase the probability that informative regions are masked, many methods employ very high masking ratios (e.g., 75% in MAE), increasing reconstruction difficulty and computational demand. By contrast, natural language has compact structure and high semantic density, enabling effective learning at comparatively low masking ratios [Devlin et al., 2019]. The high-ratio regime in image-based MIM is largely a response to visual redundancy and the absence of semantic-aware mask selection, which prevents the model from consistently allocating reconstruction capacity to critical regions.

The limitation is amplified in medical imaging because diagnostic semantics are often spatially sparse. Task-relevant evidence may be concentrated in localized lesions, organs, or anatomical subregions, whereas substantial image area corresponds to background or low-informative anatomy. Medical imaging also spans heterogeneous modalities, including MRI, CT, and X-ray, for which the geometry, scale, and spatial distribution of relevant regions vary substantially across tasks. Effective masking strategies in this setting must therefore be both semantically selective and transferable across modality-specific image statistics.

To address these requirements, we propose a controllable, text-guided masking framework, denoted **Mask-What-Matters (MWM)**. Given a user-specified open-vocabulary description or phrase, a pretrained vision-language model produces zero-shot localization cues, which are converted into robust regions of interest (ROIs). MWM then applies prompt-conditioned, region-specific masking: semantically important regions such as lesions or organs receive higher masking ratios, whereas background regions receive lower ratios. This mechanism injects downstream semantic information into the pretraining objective without requiring per-image reports or labels.

The contributions are summarized as follows:

- We formulate the first controllable text-guided masking framework for medical imaging. MWM integrates vision-language localization (BiomedCLIP) with segmentation refinement (SAM) to infer task-relevant regions from open-vocabulary prompts and to apply differentiated ROI vs. background masking, directly addressing the semantic misalignment induced by random masking.
- We provide an annotation-free and backbone-agnostic design. MWM does not require per-image reports or labels and can be integrated into ViT- and ConvNet-based MIM pipelines through a unified masking interface.
- We demonstrate consistent gains across modalities and tasks. On brain MRI, chest CT, and X-ray datasets, MWM exceeds state-of-the-art MIM baselines (e.g., SparK), improving classification, detection, and segmentation performance and supporting text-driven generative pretraining as a viable strategy for medical imaging.

2 RELATED WORK

2.1 MASKED IMAGE MODELING

Masked modeling was first popularized in natural language processing by bidirectional encoder representations from transformers (BERT) [Devlin et al., 2019], which learns contextual token representations by predicting masked tokens during pretraining. Extending this principle to vision, [He et al., 2022] introduced MIM, in which image regions are removed from the input and the model is optimized to reconstruct the missing content. This objective has shown strong generalization in both natural-image and medical-image settings [Gupta et al., 2025; Liu et al., 2021; Huang et al., 2023]. Representative instantiations include MAE [He et al., 2022], MedMAE [Gupta et al., 2025; Xiao et al., 2023], and SparK [Tian et al., 2023]. MAE uses high-ratio random masking to encourage ViTs to encode high-level semantics; MedMAE adapts the same principle to medical images through domain-specific modifications; and SparK transfers masked reconstruction to convolutional backbones by exploiting their multi-scale feature hierarchy.

Despite these advances, conventional MIM methods usually implement random or structure-agnostic masking and therefore omit explicit modeling of downstream semantic requirements. Non-selective mask sampling can allocate reconstruction pressure to irrelevant image regions and weaken transfer. Recent region-aware MIM methods attempt to improve this alignment through two primary mechanisms:

The first mechanism is loss-driven adaptive masking. Methods such as hard patches mining (HPM) [Wang et al. (2023a)] and AnatoMask [Li et al. (2024)] estimate region-level reconstruction errors and assign larger masking probabilities to high-loss areas, under the assumption that these areas contain semantically valuable structure. This criterion, however, can prioritize regions that are difficult to reconstruct but irrelevant to the target task. In brain magnetic resonance imaging (MRI), for example, complex structures such as the spinal cord or eyeballs may induce high reconstruction loss while remaining unrelated to tumor detection.

The second mechanism uses external perceptual modules to define mask candidates. Focus-MAE [Basu et al. (2024)], for instance, employs pretrained Region Proposal Networks (RPNs) to identify high-information regions for selective masking. Although this strategy can be effective for specific tasks, it depends on supervised RPN training with domain-specific annotations and therefore has limited cross-task generality.

These limitations motivate a masking mechanism that is flexible, semantically aligned, and generalizable without task-specific annotations. Natural language provides a compact and interpretable conditioning signal for specifying target semantics, making it suitable for task-aware masked modeling in an annotation-free setting.

2.2 VISION-LANGUAGE PRE-TRAINED MODELS

Recent vision-language alignment models make it increasingly practical to identify task-relevant visual regions under zero-shot and open-world conditions [Radford et al. (2021)]. The Contrastive Language-Image Pretraining (CLIP) family learns a shared image-text embedding space from large-scale paired data and has become a standard foundation for zero-shot and open-set visual recognition. In medical imaging, domain-adapted models such as BiomedCLIP [Zhang et al. (2023)] improve cross-modal alignment for biomedical image-text data and support robust downstream transfer. Although vision-language pretrained models have been widely used in supervised downstream tasks, existing studies primarily emphasize applications such as object detection and semantic segmentation [Zhong et al. (2022), Aleem et al. (2024), Koleilat et al. (2024)]. Their open-vocabulary and zero-shot localization capabilities remain underexplored as a mechanism for controlling the MIM pretraining process.

In summary, current region-aware MIM methods, including reconstruction-loss-based dynamic masking and masking guided by external perception modules, improve semantic focus only partially and continue to exhibit three limitations: dependence on additional supervision, potential mismatch between selected masks and task semantics, and limited flexibility. At the same time, vision-language models provide strong semantic localization priors, but natural language prompts have rarely been integrated into the MIM pretraining stage. These observations motivate the use of vision-language semantic guidance as a controllable interface for efficient and generalizable masked image modeling.

3 METHODOLOGY

3.1 OVERALL FRAMEWORK

As shown in Figure 1, **Mask What Matters (MWM)** consists of two coupled stages: (1) text-guided region localization and (2) region-aware masked image modeling. The localization stage uses a frozen vision-language model to infer task-relevant regions from an open-vocabulary prompt. The masked modeling stage then assigns region-dependent masking ratios and optimizes reconstruction under this semantically conditioned mask distribution.

3.2 TEXT-GUIDED REGION LOCALIZATION

MWM implements text-guided localization as a multi-stage inference pipeline. The procedure first generates a textual prompt and encodes the image-prompt pair with BiomedCLIP to obtain cross-modal embeddings. It then estimates a saliency map using the Multi-Modal Information Bottleneck (M2IB), binarizes the map with K-Means, and refines the resulting mask with the Segment Anything

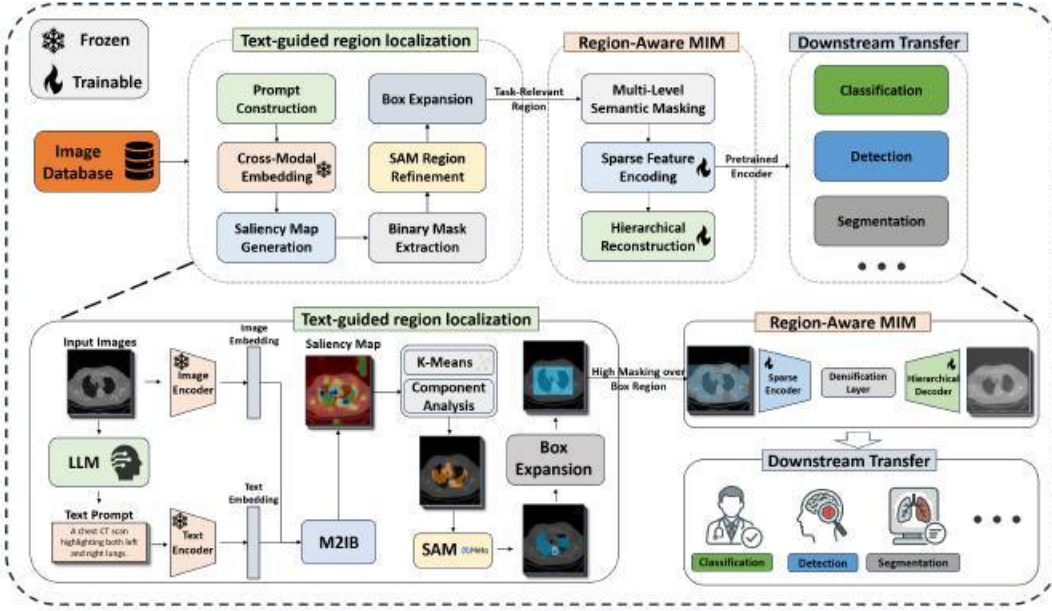


Figure 1: Schematic overview of the MWM framework. The top panel specifies the three-stage pipeline: (1) text-guided region localization, (2) region-aware masked image modeling, and (3) downstream transfer. The bottom panel details how textual prompts condition region localization and masking decisions.

Model (SAM). The refined mask is finally transformed into a bounding box with a controllable expansion margin, producing the spatial ROI used by the masked modeling stage.

Prompt Generation. In medical image analysis, prompt specification determines the semantic target used for localization. To provide descriptive guidance, large language models (LLMs) are queried with structured templates such as “Describe the typical visual characteristics of a [category] in a [modality] image”. Simple category phrases are also evaluated as lower-granularity inputs. The experimental section reports the effect of these prompt styles under matched evaluation conditions.

Cross-Modal Embedding. Given a generated prompt, BiomedCLIP [Zhang et al. \(2023\)](#) maps the medical image and text into a shared semantic embedding space. For image I and prompt T , the embeddings are computed as:

$$Z_{\text{img}} = \Phi_{\text{img}}(I), \quad Z_{\text{text}} = \Phi_{\text{text}}(T) \quad (1)$$

where Φ_{img} and Φ_{text} denote the vision and language encoders, respectively. The resulting cross-modal embeddings define the semantic alignment signal used by the subsequent saliency estimator.

Saliency Map Generation. A cross-modal saliency estimator based on the Multi-modal Information Bottleneck (M2IB) [Wang et al. \(2023b\)](#) localizes task-relevant visual regions by preserving image-text mutual information while suppressing modality-specific redundancy. Formally, the saliency map $\lambda_S \in [0, 1]^{H \times W}$ assigns an importance weight to each spatial location according to its semantic relevance to the prompt. M2IB estimates this map by optimizing the following objective:

$$\mathcal{L}_{\text{M2IB}} = \text{MI}(Z_{\text{img}}, Z_{\text{text}}) - \gamma \cdot \text{MI}(Z_{\text{img}}, I) \quad (2)$$

where Z_{img} and Z_{text} denote the BiomedCLIP image and text embeddings, respectively, and γ controls the trade-off between retaining cross-modal relevance and suppressing task-irrelevant visual information.

Binary Mask Extraction. After saliency estimation, the map is binarized with unsupervised K-Means clustering [Lloyd \(1982\)](#) to obtain a preliminary prompt-conditioned foreground mask. The map is partitioned into two pixel clusters, and the cluster with the larger saliency values is assigned to the foreground. Connected component analysis is then applied to the binary map, retaining the largest foreground components to suppress spurious edge activations and isolate the principal semantic structures.

Region Refinement. SAM is used as a spatial refinement module to improve mask boundary accuracy [Kirillov et al., 2023]. For each selected connected region $c_i \in \mathcal{C}^*$, its minimal enclosing bounding box is computed and provided to SAM as the visual prompt.

Conditioned on the original image I and the set of bounding boxes, SAM predicts the refined segmentation mask:

$$M_{\text{SAM}} = \text{SAM}(I, \text{Box}(c_i)) \quad (3)$$

The output M_{SAM} provides a spatially refined estimate of the prompt-relevant structure.

Box-Based Region Expansion. The SAM output may still contain boundary noise that could destabilize the masking distribution during representation learning. To make the region guidance more robust, the refined SAM mask is converted into a bounding box with a controllable expansion margin. This expanded box is used as the final region of interest (ROI) for downstream masked modeling.

3.3 REGION-AWARE MASKED IMAGE MODELING

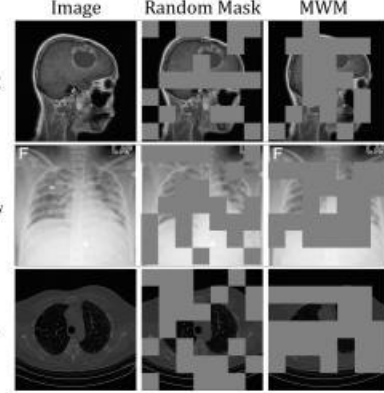
Given the ROI produced by the localization module in Section 3.2, MWM performs masked image modeling with three components: (1) a multi-level masking policy that assigns different masking ratios according to region importance, (2) a sparse encoder that processes only visible patches, and (3) a hierarchical decoder that reconstructs the full image from sparse multi-scale features.

Multi-Level Semantic Masking. MWM uses a multi-level masking policy parameterized by the semantic regions returned by text-driven localization. Patches inside prompt-highlighted regions, such as tumor- or organ-associated areas, are masked at a higher ratio so that the reconstruction objective concentrates on task-relevant semantic features. Background patches are masked at a lower ratio or kept visible, which reduces redundant computation. Figure 2 visualizes the resulting masking patterns.

Sparse Feature Encoding. Visible patches are assembled into a sparse image representation and encoded according to the SparK design [Tian et al., 2023]:

$$\begin{aligned} X_{\text{sparse}} &= \{x_i \mid m_i = 1\}, \\ F &= \Phi_{\text{sparse}}(X_{\text{sparse}}) \end{aligned} \quad (4)$$

Figure 2: Comparison of masking strategies, where gray blocks denote masked regions.



where x_i is the i -th patch, $m_i \in \{0, 1\}$ denotes its visibility indicator, Φ_{sparse} is the encoder, and $F = \{F_1, F_2, F_3, F_4\}$ represents the extracted multi-scale feature set. Although the implementation instantiates MWM with convolutional backbones, the same masking interface extends to ViTs through token indexing.

Hierarchical Reconstruction. A lightweight UNet-style decoder reconstructs the image through a top-down feature densification process. At each scale l , masked locations are replaced by a learned embedding M_l :

$$F'_l(i) = \begin{cases} F_l(i), & \text{if patch } i \text{ is visible,} \\ M_l, & \text{if patch } i \text{ is masked.} \end{cases} \quad (5)$$

The densified map F'_l is projected by the scale-specific layer ϕ_l to produce D_l , which is then up-sampled and fused with features from the next lower level:

$$D_{l-1} = B_{l-1}(D_l) + \phi_{l-1}(F'_{l-1}). \quad (6)$$

The pretraining objective minimizes mean-squared reconstruction error over masked patches:

$$\mathcal{L}_{\text{recon}} = \frac{1}{|\mathcal{I}_{\text{mask}}|} \sum_{i \in \mathcal{I}_{\text{mask}}} \|\hat{x}_i - x_i\|^2, \quad (7)$$

where $\mathcal{I}_{\text{mask}}$ denotes the set of masked patch indices, x_i is the original patch, and $\hat{x}_i$ is its reconstruction.

4 EXPERIMENTS

4.1 DATASETS

Self-Supervised Pretraining Datasets. Self-supervised pretraining uses three datasets that cover distinct medical imaging modalities. (1) *Brain MRI* includes approximately 17,000 images from Brain Tumor MRI [Cheng et al. (2015)], BRISC [Fateh et al. (2025)], and BraTS 2018 [Menze et al. (2014)], covering various tumor types and anatomical structures. (2) *Lung X-ray* consists of around 39,000 chest radiographs from COVID-QU-Ex [Rahman et al. (2021)]; Chowdhury et al. (2020), including COVID-19, non-COVID pneumonia, and normal cases. (3) *Chest CT* comprises 12,000 slices from a lung disease dataset [Konya (2020)], with segmentation masks of fibrotic lungs from 107 patients.

Downstream Fine-Tuning Datasets. Downstream evaluation is conducted on three datasets spanning classification and detection settings. (1) *Brain Tumor MRI* [Feltrin (2023)] includes 4,479 MRI images spanning 44 tumor types, such as astrocytoma, glioblastoma, meningioma, and ependymoma. (2) *Pediatric Lung X-ray* [Kermay et al. (2018)] consists of 5,863 pediatric chest X-rays categorized as either normal or pneumonia. (3) *Chest CT Cancer* [Hany (2020)] includes CT images from four classes: adenocarcinoma, large cell carcinoma, squamous cell carcinoma, and normal tissue.

4.2 EXPERIMENTAL SETUP

All self-supervised methods are configured with backbones of comparable model scale: convolutional methods use ResNet-50 [He et al. (2016)], whereas transformer-based methods use ViT-S [Dosovitskiy et al. (2020)]. Inputs are resized to 224×224, and all models are initialized from ImageNet-pretrained weights. Pretraining uses an initial learning rate of 2×10^{-4} . Downstream fine-tuning preserves the pretrained encoder and the same input resolution. Experiments are executed on NVIDIA Tesla A100 GPUs with 80 GB VRAM using Python 3.9, Ubuntu 22.04.3, and PyTorch 1.10.0. Unless otherwise specified, non-target factors such as prompt type and SAM usage are fixed to their empirically optimal settings so that each evaluation isolates the intended variable.

4.3 EVALUATION OF TEXT-GUIDED LOCALIZATION

We first quantify the text-guided localization module across brain MRI, lung X-ray, and chest CT, and then isolate the effects of prompt type and SAM refinement. Predicted bounding boxes are compared with expert-annotated segmentation masks using occlusion precision ($|R_p \cap R_{gt}|/|R_p|$) and occlusion recall ($|R_p \cap R_{gt}|/|R_{gt}|$), where R_p denotes the predicted region and R_{gt} denotes the ground-truth annotation.

Table I shows that text-guided localization with effective prompts yields consistently high occlusion recall across datasets (around 0.82) while preserving reasonable precision. This indicates that

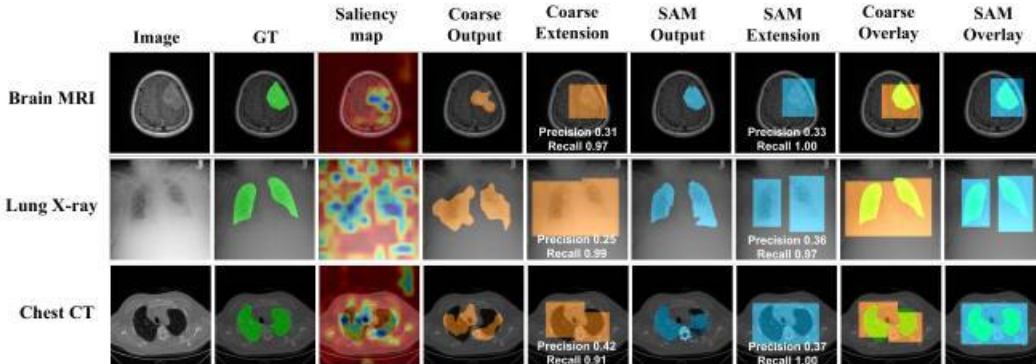


Figure 3: Representative outputs of the text-guided region localization module.

Method	Brain MRI		Chest CT		Lung X-ray	
	Prec.	Rec.	Prec.	Rec.	Prec.	Rec.
<i>Prompt Type</i>						
Phrase	0.22	0.86	0.21	0.99	0.42	0.82
Descrip. Sent.	0.30	0.94	0.37	0.88	0.44	0.66
<i>Impact of SAM</i>						
Coarse Ext.	0.22	0.95	0.36	0.87	0.37	0.84
+ SAM Ext.	0.30	0.94	0.37	0.88	0.42	0.82

Table 1: Localization performance measured by Occlusion Precision/Recall under different prompt types and refinement strategies. *Coarse Ext.* denotes bounding box extension from the coarse region; + *SAM Ext.* denotes additional refinement with SAM.

the framework can recover task-relevant spatial regions from textual descriptions alone, despite modality-specific anatomical variation; representative examples are shown in Figure 3

Effect of Prompt Type on Region Localization. To characterize the role of language granularity, we evaluate two prompt types. As reported in Table 1, descriptive sentence prompts improve localization on Brain MRI and Chest CT, suggesting that richer semantic descriptions provide more discriminative localization cues. Conversely, concise phrases achieve higher recall on lung radiographs (0.82 vs. 0.66), likely because the broad anatomical extent of the lungs makes general category terms sufficient.

Effect of SAM Refinement on Region Localization. We further evaluate spatial refinement by applying SAM to the coarse masks. Table 1 shows that SAM increases precision while maintaining high recall; for Brain MRI, precision rises from 0.22 to 0.30 while recall remains above 0.94. These results indicate that SAM improves the spatial specificity of coarse localization outputs.

4.4 COMPARISON WITH PREVIOUS SSL METHODS

We next evaluate the transferability and representation quality of MWM relative to representative self-supervised learning (SSL) methods from both reconstruction-based and contrastive paradigms. All methods are pretrained and evaluated on the same datasets with matched training settings to support controlled comparison.

Image Classification. Classification is evaluated on three downstream medical imaging datasets under two standard transfer protocols: (1) *full fine-tuning*, in which all model parameters are updated; and (2) *linear probing*, in which the encoder is frozen and only a linear classifier is optimized.

- **Full fine-tuning:** Table 2 summarizes classification accuracy for MWM and several representative self-supervised methods, including MAE He et al. (2022), SparK Tian et al. (2023), AnatoMask Li et al. (2024), SimCLR Chen et al. (2020a), MoCoV2 Chen et al. (2020b), and BYOL Grill et al. (2020). MWM achieves the highest accuracy in all three modalities, indicating robust transfer across heterogeneous medical image distributions. The largest gain occurs on

Method	Type	Brain MRI	Chest CT	Lung X-ray
MoCoV2	CL	95.3	91.7	93.1
BYOL	CL	91.3	93.7	93.9
SimCLR	CL	94.5	93.1	94.1
MAE	MIM	95.4	91.7	94.9
AnatoMask	MIM	96.5	95.8	95.2
SparK	MIM	96.2	94.4	94.7
MWM(Ours)	MIM	96.8	97.5	96.0

Table 2: Downstream fine-tuning classification accuracy (%). MIM denotes Masked Image Modeling, and CL denotes Contrastive Learning.

Method	Brain MRI	Chest CT	Lung X-ray
AnatoMask	70.2	63.9	85.3
SparK	69.8	63.5	81.6
MWM(Ours)	71.3	67.0	88.5

Table 3: Downstream linear probing accuracy (%) with the encoder frozen.

Method	Detection		Segmentation	
	AP ^{box}	AP ^{box} ₇₅	AP ^{mask}	AP ^{mask} ₇₅
AnatoMask	45.2	53.1	46.2	54.1
SparK	45.6	53.2	44.8	52.0
MWM (Ours)	46.9	53.5	45.9	54.1

Table 4: BR35H detection and instance segmentation performance (%), reported as AP^{box}, AP^{box}₇₅, AP^{mask}, and AP^{mask}₇₅.

Chest CT, where MWM obtains 97.5% accuracy, exceeding SparK by +3.1 and AnatoMask by +1.7. This improvement is consistent with the intended effect of text-guided masking: reconstruction pressure is concentrated on semantically critical but spatially sparse regions, such as lung lobes in CT scans, while redundant background masking is reduced.

- **Linear probing:** Linear probing further assesses whether the frozen encoder contains transferable features. We compare MWM with two recent masked image modeling methods, SparK and AnatoMask. As shown in Table 3, MWM outperforms both baselines on all datasets, indicating that text-driven pretraining improves the separability of downstream representations without encoder adaptation.

Object Detection and Instance Segmentation. To test transfer beyond classification, we evaluate brain tumor detection and instance segmentation on BR35H [Hamada (2025)]. Table 4 reports that MWM obtains the highest AP^{box} (46.9%) and the second-highest AP^{mask} (45.9%), exceeding SparK by +1.3 and +1.1 points, respectively. These results show that the learned representations transfer to both object-level localization and pixel-wise segmentation.

Overall, text-driven masking during pretraining yields representations with stronger downstream transfer. The consistent improvements across classification, detection, and segmentation indicate that semantic control over the masking distribution enhances visual representation learning.

4.5 ABLATION STUDY

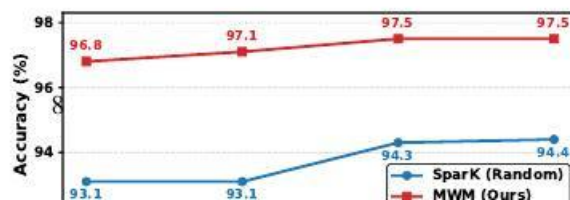
Ablation experiments isolate the contributions of prompt design, SAM-based region refinement, and masking ratio to MWM downstream classification performance.

4.5.1 EFFECT OF PROMPT DESIGN AND REGION REFINEMENT

As shown in Table 5, descriptive sentence prompts outperform short phrases on brain MRI and chest CT, whereas short phrases perform better on lung X-ray. This pattern is consistent with the localization analysis in Section 4.3: prompt formulations that produce more accurate region localization also yield stronger downstream accuracy. Both prompt types, independent of granularity, outperform random masking on all datasets. This result supports the claim that text-guided masking improves semantic alignment between the pretraining objective and downstream task structure. The SAM ablation further shows that incorporating spatial refinement during pretraining consistently improves downstream classification, indicating that more precise region guidance strengthens text-conditioned masking.

4.5.2 EFFECT OF MASKING RATIO

As shown in Figure 4, MWM reaches 96.8% classification accuracy with a 40%



Group	Setting	Brain MRI	Chest CT	Lung X-ray
Prompt Type	No Prompt	96.2	94.4	94.7
	Phrase	96.4	96.5	96.0
	Sentence	96.8	97.5	95.2
Impact of SAM	w/o SAM	96.4	96.8	95.8
	w/ SAM	96.8	97.5	96.0

Table 5: Downstream classification accuracy (%) for prompt-type and region-refinement ablations across three imaging modalities. *No Prompt* denotes random masking without text-guided localization. *w/* and *w/o* denote with and without SAM, respectively.

masking ratio, which is substantially lower than the approximately 70% ratio commonly used in masked image modeling. Even with this reduced ratio, MWM exceeds SparK’s best result at 70% masking by +2.4% absolute accuracy. To exclude the possibility that SparK is disadvantaged by a suboptimal mask ratio, SparK is additionally evaluated at 40%, 50%, 60%, and 70% masking. The results show that 70% is the strongest configuration for SparK, supporting the validity of the comparison.

These findings qualify the assumption that high masking ratios are intrinsically optimal for MIM. Instead, they indicate that semantically guided, task-aware masking can improve downstream performance at lower masking ratios by preserving task-relevant context and reducing redundant reconstruction.

5 CONCLUSION AND LIMITATIONS

This work introduced **Mask What Matters (MWM)**, a self-supervised pretraining framework that combines text-guided semantic localization with region-aware masking. By using natural language prompts to specify task-relevant regions and applying differentiated masking ratios, MWM aligns the reconstruction objective with downstream semantics without requiring per-image annotations.

The experimental results support three conclusions: (1) natural language prompts provide reliable cues for semantic region localization, validating text-driven masking as a practical control mechanism; (2) MWM outperforms existing methods (e.g., SparK, AnatoMask) for classification across three imaging modalities and also improves detection and instance segmentation metrics; (3) MWM remains effective at lower masking ratios (e.g., 40%), indicating that semantic guidance can reduce the need for high-ratio random masking in self-supervised pretraining.

Limitations. The framework has two primary limitations. First, although text prompts provide a flexible control interface, the robustness of MWM to prompt style, vocabulary variation, and textual noise has not been systematically evaluated. Second, the use of external textual descriptions introduces a dependency that may reduce generalization when reliable prompts are unavailable or inconsistent.

Future work should examine text-driven, region-aware masking as a general mechanism for semantically grounded self-supervised learning and evaluate whether the MWM formulation transfers beyond medical imaging to broader vision–language pretraining settings.

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A USE OF LARGE LANGUAGE MODELS (LLMs)

LLMs were used *only* for language editing and presentation polishing of author-written text, including grammar, phrasing, and minor stylistic clarification. LLMs were *not* used for idea generation, experimental design, data annotation, code implementation, or analysis/interpretation of results. All technical content, algorithms, proofs, figures, tables, metrics, and conclusions were authored and verified by the authors. No confidential or identifying data were provided to LLMs beyond anonymized manuscript text.